



NATIONAL HARMONISED PROGRESSION SHEET FOR COMPUTER SCIENCE FORM 2

SCHOOL YEAR.....

SCHOOL.....

WEEKLY WORKLOAD: 2 periods

TEACHER.....

GRADE.....

TEL.....

Term	Week	Module	Category of action	Competency statement	Lesson title	Objectives	Resources	Nature of lesson		
								Th	Prac	Dig
FIRST TERM	1	Computing environment 2	Classifying hardware components	Given a set of hardware, learners classify devices into appropriate categories	Diagnostic evaluation Lesson 1: Types of Input and output devices	- State the main role of input and output devices. - Identify input and output devices.				
					Lesson 2: Types of processing devices and storage devices	- State the function of a given processing device. - State the function of a given storage device. - Identify processing and storage devices using AI simulation.	Use AI Simulation where appropriate			
					Lesson 3: Integration activities					
	2		Exploring Application Areas of Computers	Provided with a situation related to the use of AI, learners write appropriate prompts to get an output or response.	Lesson 4: Introduction to Artificial Intelligence	- Define AI - Explain the history and evolution of AI - Identify and categorise the different types of AI in real world	Use AI Simulation where appropriate			
					Lesson 5: Applications and Ethics of Artificial Intelligence	- Identify common applications of AI - Explain the uses of AI in different fields - State the ethical issues related to AI	Use AI Simulation where appropriate			
					Lesson 6: The use of appropriate Prompts to generate AI responses	- Examine prompt results - Discuss prompt results - Write appropriate prompts to solve problems within a given context.	Debate, Simulation			
					Lesson 7: Integration activities					
	3									
4										

		Computational thinking and coding	Problem solving strategies	Learners develop strategies to solve problems in a logical manner	Lesson 8: Solving problems 1	• Apply the steps of problem solving to a problem. • Apply decomposition and pattern recognition to a problem using AI simulation. • Differentiate 2D representations from 3D	Use AI Simulation where appropriate					
	5				Lesson 9: Solving problems 2	• Apply abstraction to a problem. • Write the solution to a problem as a sequence of steps such that key characteristics of abstraction are seen in the solution. • Implement it using scratch	Use Scratch Programming where appropriate					
					Lesson 10: Integration activities							
					Evaluation							
	Remediation											
	Computing environment 2	7	Choosing appropriate system software	Given a situation, learners select system software that is in line with the situation.	Lesson 11: Notions on operating systems	• Explain the concept of operating systems. • State common types of operating systems and give examples of each. • Outline the functions of an operating system.						
					Lesson 12: Notions on utility software and device drivers	• Differentiate between utility software and device drivers. • State the function of a given type of utility software with examples. • State the purpose of device drivers.						
					Lesson 13: Integration activities							
		8	Representing numbers and characters	Given a situation in relation to encoding data, learners apply the right encoding technique to represent numbers and characters correctly.	Lesson 14: Introduction to number systems	• State the number system used in computing • Explain how to count in base 2, 8, and 16 • Convert from base 2 to base 10						
					Lesson 15: Notions on encoding numbers	• Explain the concept of encoding, and number encoding. • Explain the technique used to encode positive integers. • Convert from base 10 to base 2.						
					Lesson 16: Notions on encoding characters	• Explain the concept of character encoding • State character encoding schemes • Make use of the ASCII character encoding scheme.						
		Lesson 17: Integration activities										
		9			Converting between units of time and storage	Given time units or storage units, learners compare and convert from one unit to another.	Lesson 18: Conversion between units of time	• Identify units of time. • Arrange units of time in a particular order. • Convert between units of time.				
							Lesson 19: Conversion between units of storage	• Identify units of storage. • Arrange units of storage in a particular order. • Convert from one unit of storage to another.				
Lesson 20: Integration												

	12				activities					
					Evaluation					
					Remediation					
SECOND TERM	13	Computing environment 2	Exploring the internet and browsers	Provided with a situation related to the access and use of the internet, learners assemble the necessary and adapted tools needed to access the internet and use basic features of a browser correctly.	Lesson 21: Accessing the internet	- State the components needed to access the internet. - Outline elements influencing the growth of the internet. - Choose appropriate ISP in a given context.				
					Lesson 22: Using a browser	- Explain the concepts of tabs, browsing history, and bookmark. - Identify common features on the interface of a browser. - Perform basic operations that can be done with a browser.				
					Lesson 23: Integration activities					
	14	Examining technologies for the disabled	Provided with a situation with factors related to users with disabilities, learners identify appropriate technologies for a given disability within a context	Lesson 24: Assistive technologies	- Explain the concepts of disability, and assistive technology - Match a given assistive technology for an appropriate disability using AI simulation. - State common assistive technologies integrated in an operating system	Use AI Simulation where appropriate				
					Lesson 25: Integration activities					
	15	Computational thinking and coding 2	Notions of programming	Given a project or problem related to coding, learners with the guidance of the teachers use features of a block programming tool to produce code that responds to the project or situation.	Lesson 26: Writing programs in a block programming environment 1	- Work in block programming environments. - Write programs using a block programming environment.				
					Lesson 27: Writing programs in a block programming environment 2	Write programs using a block programming environment				
					Lesson 28: Integration activities					
	16	Computational thinking and coding 2	Notions of programming	Given a project or problem related to coding, learners with the guidance of the teachers use features of a block programming tool to produce code that responds to the project or situation.	Lesson 29: Notions on digital Identity	- Explain the concepts of digital identity, and identity theft. - Outline common components of a digital identity (username and password). - Outline best practices to manage multiple digital identities.				
					Lesson 30: Types of digital footprint	- Explain the concept of digital footprints. - State ways users create digital footprints. - Differentiate between passive and active digital				
	17	Digital citizenship 2	Managing digital identities	Provided with a situation with issues related to digital identities, digital footprints, and reputation, learners recommend best practices to deal with or prevent such						

				a situation. The practices should be pertinent and coherent to the situation.		footprints.				
	18				Evaluation					
					Remediation					
	19	Computing environment 2	Understanding embedded systems, IoT and cloud computing	Given a situation with factors related to IoT and embedded systems, learners identify the correct type of sensors used in the situation and their role.	Lesson 31: Managing digital footprints	- State positive and negative effects of digital footprints. - Outline ways of minimising digital footprints. - Discuss ways of protecting online reputation.				
					Lesson 32: Initiation to cloud computing	- Define cloud computing. - Outline the 3 main types of cloud computing services. - Outline examples of cloud services.				
	20				Lesson 33: Types of sensors	- State common types of sensors. - State the function of a given type of sensor. - Differentiate between a sensor and an actuator.				
					Lesson 34: Notions on embedded systems and IOT	- Differentiate between embedded systems and IoT. - Describe situations where IoT and embedded systems are used. - Explain how IoT, embedded systems and cloud computing work together.				
	21				Lesson 35: Integration activities					
		Computing environment 2	Using presentation, desktop publishing and spreadsheet software	Given a task that requires desktop publishing, spreadsheet software or presentation software, learners choose adapted software and produce an output that corresponds to the task given.	Lesson 36: Notions on Spreadsheet software	- State common examples of spreadsheet software. - Outline core features of a spreadsheet software. - Identify when to use an electronic spreadsheet.				
					Lesson 37: Using a spreadsheet to solve problems	- Carry out simple formatting using a spreadsheet. - Perform simple calculations using a spreadsheet.				
	22				Lesson 38: Notions on presentation software and desktop publishing software.	- State common examples of presentation and desktop publishing software. - Outline core features of presentation and desktop publishing software. - Identify when to use presentation and desktop publishing software.				
					Lesson 39: Using presentation software to solve problems	- Carry out simple formatting using presentation software. - Produce slides with different layouts.				
	23				Lesson 40: Using desktop publishing software to solve problems	- Carry out simple formatting using desktop publishing software. - Produce simple publications using desktop publishing software.				
					Lesson 41: Integration activities					
	24				Evaluation					

THIRD TERM	25		Protecting against data loss and unauthorised access	Presented with a situation with factors related to protecting data, learners propose appropriate protection mechanisms to fight against loss and unauthorised access.	Remediation					
					Lesson 42: Notions on authentication	- Explain the concept of unauthorised access, and authentication. - Outline authentication techniques used to protect data from unauthorised access.				
					Lesson 43: Notions on backup	- Explain the concepts of data loss, backup, and recovery. - Explain how backup prevents data loss. - Compare local backup and cloud backup based on storage location, accessibility, and security.				
	26				Lesson 44: Integration activity					
	27	Computing environment 2	Using communication and collaboration tools	Given a situation that requires the use of software to perform a task, learners create expected content using appropriate software.	Lesson 45: Communication and collaboration tools	- Differentiate between communication and collaboration. - Give examples of communication tools and collaboration tools. - Give examples and make use of cloud word processors to collaborate.	Use AI Simulation where appropriate			
					Lesson 46: Email tool	- Create an email account in a given email service. - Make use of an email service to send mails, read mails, and attach files.				
					Lesson 47: Chat tool	- Make use of a chat tool to send and receive text messages. - Make use of a chat tool to send non-text messages. - Make use of a chat tool to respond to a specific message in a conversation.	Use AI Simulation where appropriate			
					Lesson 48: Videoconference tool	- Create a videoconference or meeting using an appropriate tool. - Make appropriate use of basic features in a videoconference tool.	Use AI Simulation where appropriate			
					Lesson 49: Integration activities					
					Evaluation					
					Remediation					
	28				Lesson 50: introduction to computer maintenance	- Explain the concept of computer maintenance. - Classify computer faults into hardware and software issues. - Differentiate between preventive and corrective maintenance.				
					Lesson 51: Common hardware problems	- Identify the key characteristics of a given hardware problem. - State a possible solution to a given hardware problem. - Match a given hardware problem to a measure to				
29	30	Computing environment 2	Applying basic preventive and corrective maintenance	Given a situation presenting computer faults, learners propose basic corrective and preventive maintenance actions to solve or stop such issues. The action proposed should be						
31										

				logical and pertinent.		prevent it.					
					Lesson 52: Common software problems	- Identify the key characteristics of a given software problem. - State a possible solution to a given software problem. - Match a given software problem to a measure to prevent it.					
					Lesson 53: Integration activities						
	32	Digital citizenship 2	Evaluating the credibility and reliability of sources of information	Given a problem with factors related to the credibility of information, learners detect the credibility and reliability of common sources of information and digital content and recommend trustworthy sources to access information and digital content	Lesson 54: Verifying the credibility of information	- Differentiate between primary and secondary sources of information - Explain ways of verifying the credibility of information - State trustworthy sources of information and digital content					
	33				Lesson 55: Integration activities						
					Evaluation						
	34				Remediation						
			Revision								
	35		END OF PROGRAM								
	36										